Strong and stringent hamstring strain science: trials and error!

Kristian Thorborg, Lasse Ishøi, Kasper Krommes

Knowledge on how to optimise rehabilitation of athletes after an acute hamstring strain is progressing.[1] Within the last 6 years, eight randomised controlled trials (RCTs) on this topic have emerged—conducted by many eminent clinician-scientists.[1]

FROM CLINICAL QUESTION TO SCIENCE!

Recent clinically relevant RCTs show that specific exercise-based interventions reduce time to return to sport and risk of re-injuries.[1] In particular, lengthening contraction exercises compared with conventional exercises without such emphasis reduce time to return to sport,[1] whereas a criteria-based programme including a more running-specific exercise focus, compared with the lengthening approach, seems promising for reducing the risk of re-injury.[1] Furthermore, platelet-rich plasma injections provided no additional benefit to current exercise-based rehabilitation.[1] This emphasises the importance of specific mechanical loading and progressive running exercises in athletes with hamstring strains—but can we rest on these laurels?

This editorial points out common methodological shortcomings in the existing clinical hamstring trial literature that may help guide and improve scientific robustness of future trials. Based on these, we suggest four practical steps to advance the field by eliminating common errors:

> "The aim of science is not to open the door to infinite wisdom, but to set a limit to infinite error". German dramatist Bertolt Brecht (1898–1956)

Step 1—reporting guidelines

Follow established preregistration practices, use rigorous study designs and follow reporting guidelines such as Consolidated Standards of Reporting Trials (RCTs) and Consensus on Exercise Reporting Template (exercise trials).[2] A priori defined hypotheses, analyses and outcomes must become the new norm. As an example of the current state of affairs only 2[3 4] out of the 10 existing hamstring strain rehabilitation RCTs were preregistered. Of those, neither study followed the registered outcome reporting plan, nor reported or justified these deviations.[3 4]

Future hamstring trials can also be improved by using the PICOT (Population, Intervention, Comparator, Outcome and Time of interest) approach[2] and we provide specific tips for each of those elements in the subsequent steps.

Step 2—population, intervention and comparator

Define and investigate a homogeneous group of specific athletes with specific injury severity, with more attention directed towards sex, level of play and performance. The (P)opulations studied in existing trials have mainly been male non-professional athletes from mixed sports.[1] The athletes present with a wide range of injury severity (from only a few days to many months of recovery).[1] While most recent trials suggest return to sport to average at >3 weeks,[1] professional football players return much quicker.[5 6] As elite athletes, such as footballers, have the highest risk of hamstring re-injury during the first weeks after rehabilitation,[6] it is not likely that the low re-injury rates (0%–10%)[1] seen in many of the hamstring strain rehabilitation trials that include recreational athletes can be expected in professional football where re-injury rates are usually much higher (15%–30%).[5 6]

Describe the exercise (I)nterventions and (C)omparisons applied in the existing hamstring strain rehabilitation trials according to existing reporting guidelines.[2] Most interventions include a progressive running approach, but most trials do not sufficiently describe this part. As running—especially sprinting—is a very hamstring-dominant 'exercise' (lengthening contractions at high velocity),[7] we recommend describing the exact exposure to this specific stimulus, as this will permit other practitioners to replicate the specific hamstring loads applied in any given study if they so desire. We advocate that future trials report the adherence with both the main exercise-based intervention and the additional progressive running programme.

Step 3—outcome

Carefully consider the most relevant outcome for your group of athletes. The (O)utcome most commonly chosen in hamstring strain rehabilitation trials have been time to return to sport, and re-injuries. These outcomes have, however, always been assumed to be of highest relevance without considering athlete/patient input.[8] The definitions of these outcomes have not been well standardised, which limits trial interpretation and comparisons. Ten published hamstring trials use four different return to sport definitions: (1) Rehab-length. 2) Criteria. 3) Self-report. 4) Criteria and self-report, and two different follow-up time-periods (6 months and 12 months) for re-injury (figure 1). We advise that researchers in the future report the time from injury to: (1) Criteria clearance or clearance to return to sport. (2) Actual return to sport (self-reported) separately.

Step 4—time

The nature and the (T)ime period of interest concerning re-injuries are also extremely relevant to report.[2] The studies published to date did not distinguish between minor exacerbations (during rehabilitation of the original injury—before return to sport) and frank recurrences (after return to sport). Follow-up times to capture exacerbations and recurrences also vary from study to study (figure 1).

We hope these four steps taken together will advance future individual hamstring trials and facilitate the work of those who try to conduct quality meta-analysis (ideally with individual patient data). Such studies will help to set a limit to infinite error when athletes with hamstring strains rehabilitate.

Sports Orthopaedic Research Center–Copenhagen (SORC-C), Arthroscopic Center, Department of Orthopedic Surgery, Copenhagen University Hospital, Amager-Hvidovre, Denmark, Amager-Hvidovre Hospital, Hvidovre, Copenhagen, Denmark

Correspondence to Prof Kristian Thorborg, Sports Orthopaedic Research Center–Copenhagen (SORC-C), Arthroscopic Center, Department of Orthopedic Surgery, Copenhagen University Hospital, Amager-Hvidovre, Denmark, Amager-Hvidovre Hospital, Hvidovre 2650, Denmark; kristianthorborg@hotmail.com

Twitter Kristian Thorborg @KThorborg, Lasse Ishøi @Lasselshoei and Kasper Krommes @krommes

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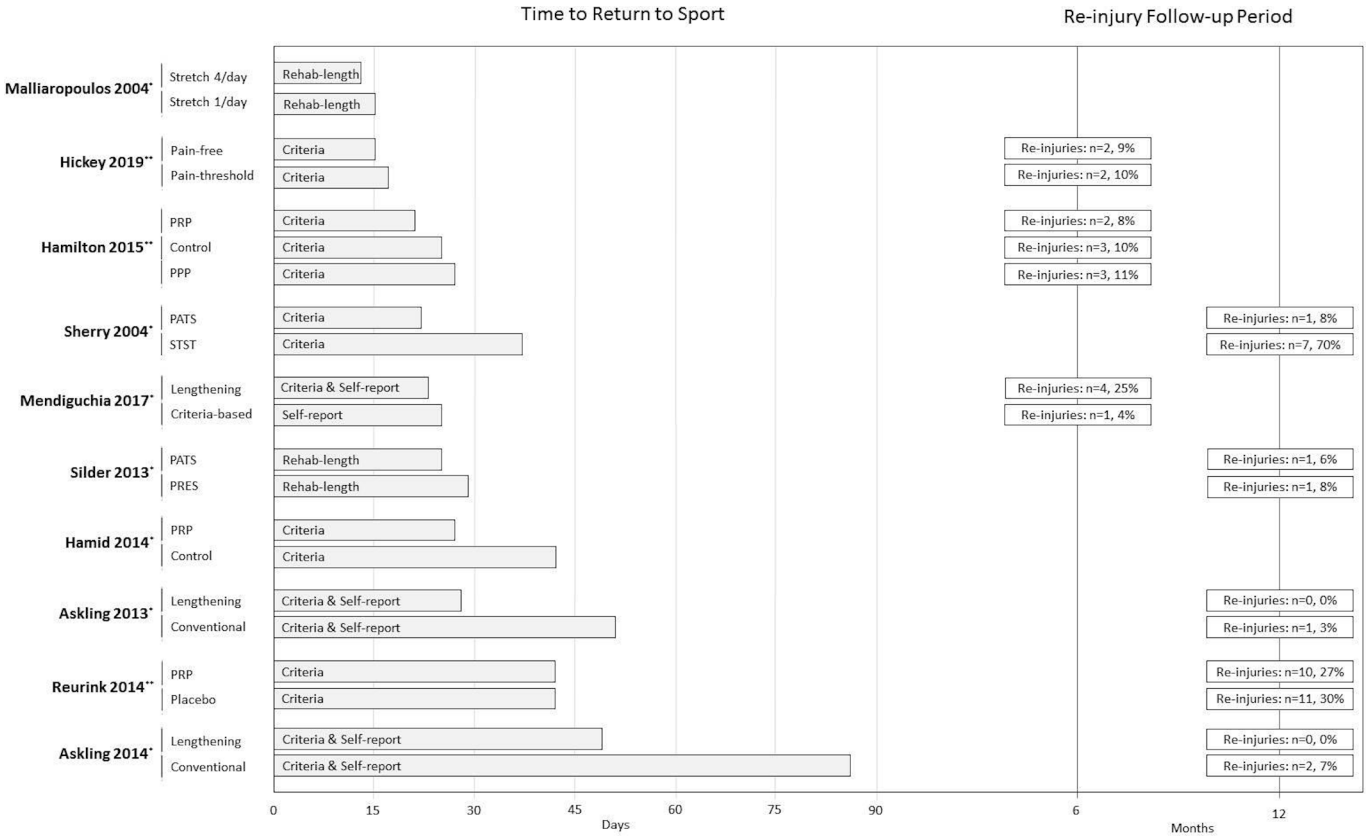

Figure 1 Time to return to sport and re-injury follow-up period in hamstring strain rehabilitation trials. Rehab-length: return to sport was considered when athletes finished their rehabilitation; Criteria: return to sport was considered when athletes passed certain criteria; Self-report: return to sport was considered when a return was reported by the athlete. * Denotes time to return to sport as the mean time; ** Denotes time to return to sport as the median time. PATS, progressive agility and trunk stabilisation; PPP, platelet-poor plasma injections; PRES, progressive running and eccentric strengthening; PRP, platelet-rich plasma injections; STST, static stretching and strengthening.

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ORCID iDs
Kristian Thorborg http://orcid.org/0000-0001-9102-4515
Lasse Ishøi http://orcid.org/0000-0002-2716-6567
Kasper Krommes http://orcid.org/0000-0001-9856-7547

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